decision-tree-1

May 10, 2025

[5]:
```
!pip install pandas
```

```
Collecting pandas
  Using cached pandas-2.2.3-cp312-cp312-macosx_11_0_arm64.whl.metadata (89 kB)
Requirement already satisfied: numpy>=1.26.0 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
pandas) (2.2.4)
Requirement already satisfied: python-dateutil>=2.8.2 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
pandas) (2.9.0.post0)
Collecting pytz>=2020.1 (from pandas)
  Downloading pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas)
  Downloading tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Requirement already satisfied: six>=1.5 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
python-dateutil>=2.8.2->pandas) (1.16.0)
Downloading pandas-2.2.3-cp312-cp312-macosx_11_0_arm64.whl (11.4 MB)
                         11.4/11.4 MB
22.9 MB/s eta 0:00:00a 0:00:01
Downloading pytz-2025.2-py2.py3-none-any.whl (509 kB)
Downloading tzdata-2025.2-py2.py3-none-any.whl (347 kB)
Installing collected packages: pytz, tzdata, pandas
Successfully installed pandas-2.2.3 pytz-2025.2 tzdata-2025.2
```

[19]:
```python
# Step 1: Import Libraries
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Step 2: Load the Dataset
data = pd.read_csv('bill_authentication.csv')

# Step 3: Prepare Features and Labels
X = data.drop('Class', axis=1)
y = data['Class']
```

```python
# Step 4: Split Data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

# Step 5: Train the Model
clf = DecisionTreeClassifier()
clf.fit(X_train, y_train)

# Step 6: Predict and Evaluate
y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy}")
```

```
Accuracy: 0.9818181818181818
```

[21]:
```python
dtree = pd.read_csv('bill_authentication.csv')
dtree.head()
```

[21]:

	Variance	Skewness	Curtosis	Entropy	Class
0	3.62160	8.6661	-2.8073	-0.44699	0
1	4.54590	8.1674	-2.4586	-1.46210	0
2	3.86600	-2.6383	1.9242	0.10645	0
3	3.45660	9.5228	-4.0112	-3.59440	0
4	0.32924	-4.4552	4.5718	-0.98880	0

[23]:
```python
X = dtree.drop('Class', axis=1)
y = dtree['Class']
```

[50]:
```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.15)
```

[51]:
```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.40)
```

[52]:
```python
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20)
```

[53]:
```python
from sklearn.tree import DecisionTreeClassifier
clf_entropy = DecisionTreeClassifier(max_depth=2)
clf_entropy.fit(X_train,y_train)

#print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
```

[53]:
```
DecisionTreeClassifier(max_depth=2)
```

[66]:
```python
y_pred = clf_entropy.predict(X_test)
from sklearn.metrics import classification_report, confusion_matrix
```

```python
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
y_pred = clf_entropy.predict([[3.45660,9.5228,-4.0112,-3.59440]])
print(y_pred)
```

```
[[137  11]
 [ 22 105]]
              precision    recall  f1-score   support

           0       0.86      0.93      0.89       148
           1       0.91      0.83      0.86       127

    accuracy                           0.88       275
   macro avg       0.88      0.88      0.88       275
weighted avg       0.88      0.88      0.88       275

[0]

/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-
packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid
feature names, but DecisionTreeClassifier was fitted with feature names
  warnings.warn(
```

[67]:
```
pip install matplotlib
```

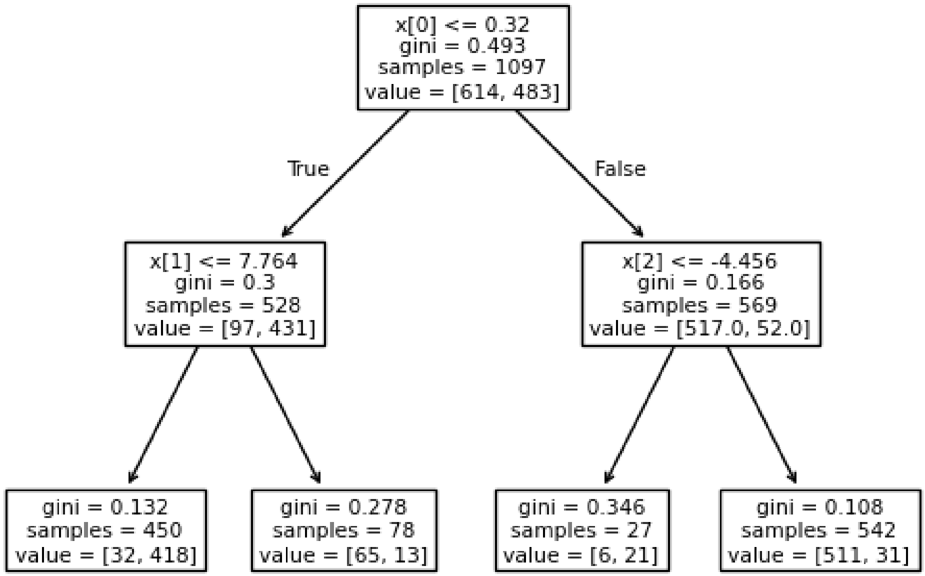
```
Requirement already satisfied: matplotlib in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (3.10.1)
Requirement already satisfied: contourpy>=1.0.1 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (4.57.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (1.4.8)
Requirement already satisfied: numpy>=1.23 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (2.2.4)
Requirement already satisfied: packaging>=20.0 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (24.2)
Requirement already satisfied: pillow>=8 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (11.2.1)
```

```
Requirement already satisfied: pyparsing>=2.3.1 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-packages (from
python-dateutil>=2.7->matplotlib) (1.16.0)
Note: you may need to restart the kernel to use updated packages.
```

[68]:
```python
import matplotlib.pyplot as plt
```

[69]:
```python
from sklearn import tree
tree.plot_tree(clf_entropy.fit(X_train,y_train))
```

[69]:
```
[Text(0.5, 0.8333333333333334, 'x[0] <= 0.32\ngini = 0.493\nsamples =
1097\nvalue = [614, 483]'),
 Text(0.25, 0.5, 'x[1] <= 7.764\ngini = 0.3\nsamples = 528\nvalue = [97, 431]'),
 Text(0.375, 0.6666666666666667, 'True  '),
 Text(0.125, 0.16666666666666666, 'gini = 0.132\nsamples = 450\nvalue = [32,
418]'),
 Text(0.375, 0.16666666666666666, 'gini = 0.278\nsamples = 78\nvalue = [65,
13]'),
 Text(0.75, 0.5, 'x[2] <= -4.456\ngini = 0.166\nsamples = 569\nvalue = [517.0,
52.0]'),
 Text(0.625, 0.6666666666666667, '  False'),
 Text(0.625, 0.16666666666666666, 'gini = 0.346\nsamples = 27\nvalue = [6,
21]'),
 Text(0.875, 0.16666666666666666, 'gini = 0.108\nsamples = 542\nvalue = [511,
31]')]
```



[70]:

```
test_size=[0.15,0.40,0.20]
results={}
for size in test_size:
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=size,␣
 ↪random_state=42)
    clf = DecisionTreeClassifier()
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    results[size] = accuracy
    print(f"Test size {size}: Accuracy = {accuracy:.4f}")

# Identify best test size
best_size = max(results, key=results.get)
print(f"\n Best test_size: {best_size} with accuracy {results[best_size]:.4f}")
```

```
Test size 0.15: Accuracy = 0.9806
Test size 0.4: Accuracy = 0.9818
Test size 0.2: Accuracy = 0.9782

 Best test_size: 0.4 with accuracy 0.9818
```

[79]:
```
from sklearn.tree import DecisionTreeClassifier
clf_entropy = DecisionTreeClassifier(max_depth=1)
clf_entropy.fit(X_train,y_train)

#print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
```

[79]:
```
DecisionTreeClassifier(max_depth=1)
```

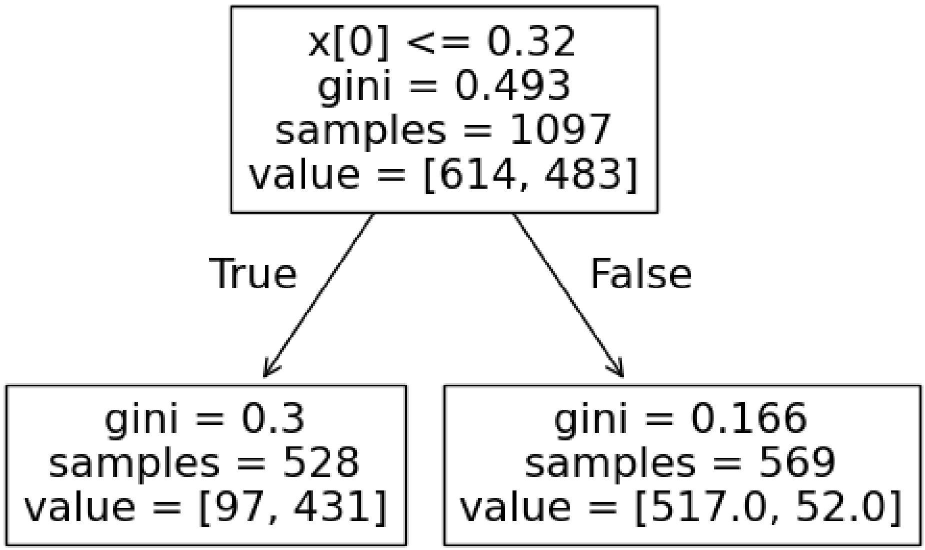
[80]:
```
y_pred = clf_entropy.predict(X_test)
from sklearn.metrics import classification_report, confusion_matrix
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
y_pred = clf_entropy.predict([[3.86600,-2.6383,1.9242,0.10645]])
print(y_pred)
```

```
[[121  27]
 [ 25 102]]
              precision    recall  f1-score   support

           0       0.83      0.82      0.82       148
           1       0.79      0.80      0.80       127

    accuracy                           0.81       275
   macro avg       0.81      0.81      0.81       275
weighted avg       0.81      0.81      0.81       275

[0]
```

```
/Users/rajthakur/anaconda3/envs/Thakur_ML/lib/python3.12/site-
packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid
feature names, but DecisionTreeClassifier was fitted with feature names
  warnings.warn(
```

[81]:
```
from sklearn import tree
tree.plot_tree(clf_entropy.fit(X_train,y_train))
```

[81]:
```
[Text(0.5, 0.75, 'x[0] <= 0.32\ngini = 0.493\nsamples = 1097\nvalue = [614,
483]'),
 Text(0.25, 0.25, 'gini = 0.3\nsamples = 528\nvalue = [97, 431]'),
 Text(0.375, 0.5, 'True  '),
 Text(0.75, 0.25, 'gini = 0.166\nsamples = 569\nvalue = [517.0, 52.0]'),
 Text(0.625, 0.5, '  False')]
```


x[0] <= 0.32
gini = 0.493
samples = 1097
value = [614, 483]
True
False
gini = 0.3
samples = 528
value = [97, 431]
gini = 0.166
samples = 569
value = [517.0, 52.0]

[ ]: